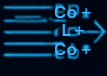


Network Detection & Response (NDR) Platform

System Architecture

01. TRAFFIC SENSING & INGESTION

Raw Traffic &
Replayed
PCAPS



Zeek Sensor

conn. log, dms. log,
\$1. log (J43), http. log



Suricata Sensor

eve. json from alerts

02. TABULAR FEATURE ENGINEERING

NormalizedFlow Common Schema



31+ Flow
Features (Rates,
Ratios, TCF Flags)



Shannon Payload &
DNS
SubdomainEntropy



Destination Port
DistributionEntropy



Packet Inter-Arrival
Timiny & Burstiness
Animer Stats — \$10

03. DUAL-PATH HYBRID DETECTION



Suricata
Signature Engine

Emerging Threats Rules
& MITRE SIDS



Supervised
FlowClassifier

XGBoost / LightGBM
Precision-First
Cutoff = 0.85



PyTorch Benign
Autoencoder

Unsupervised Zero-Day
Anomaly Detection via
NCT Reconstruction Loss

04. FAIL-SECURE DECISION ARBITER



Decision Engine
@fail_secure_guard
Watchdog
@fail_secure_guard
Watchdog



[CRITICAL /
TIMEOUT / ERROR]:
BLOAN / RATE
BLESSING & DATE IT
DESTRUCTION / BLOCK
ON CRASH OR TIMEOUT



[ANOMALOUS / UNSEEN]:
QUARANTINE VLAN 99



[BENIGN / LOW
RISK]: PASS

05. ACTIVE CONTAINMENT & VISIBILITY

ContainmentManager Engine



PRIMARY ACTION: ACTXEN
OPhense Firewall REST API
(apis/firewall/alias_util/add)



FAILURE SAFETY
Local Host Iptables
Packet Drop Rule



SIEM Dispatcher



Kezu / ELK SOC Security
(Desktop (ECS Jason Format))



99.98%

Precision | 0.04% FPR



100,000+

Real Network
Flows Evaluated



100%

Fail Secure Automated
Containment